

ACCTG 211H — Mechanics Drill Set

Week 12 — Variances, ROI, Residual Income, and EVA

Drill Set

Required: 1–2 and 5–6. Optional: 3–4 and 7. Complete the required items after instruction by **April 11, 2027, 11:59 p.m. Eastern**. Work individually by hand, check the key, and correct errors. Graded for completion; optional items are excluded from quizzes and the final.

Label variances favorable (F) or unfavorable (U). Items 1 and 2 use different quantities; do not add them as one total usage-cost variance.

1. (*Materials price variance.*) The standard price is \$5.00 per pound; the actual price is \$5.30 per pound; the actual quantity purchased is 8,000 pounds. Compute the direct-materials price variance.
2. (*Materials quantity variance.*) The standard is 3 pounds per unit at \$5.00 per pound; 2,000 units are produced; actual usage is 6,300 pounds. Compute the direct-materials quantity variance.
3. (*Labor rate variance.*) The standard rate is \$20.00 per hour; the actual rate is \$19.50 per hour; actual hours worked total 4,200. Compute the direct-labor rate variance.
4. (*Labor efficiency variance.*) The standard is 2 hours per unit at \$20.00 per hour; 2,000 units are produced; actual hours total 4,200. Compute the direct-labor efficiency variance.
5. (*Return on investment.*) A division reports operating income of \$600,000 and average operating assets of \$3,000,000. Compute ROI.
6. (*Residual income.*) For the same division (operating income \$600,000; operating assets \$3,000,000), the required rate of return is 14%. Compute residual income.
7. (*Economic value added.*) A firm has NOPAT of \$1,200,000, invested capital of \$8,000,000, and a weighted-average cost of capital of 10%. Compute EVA.